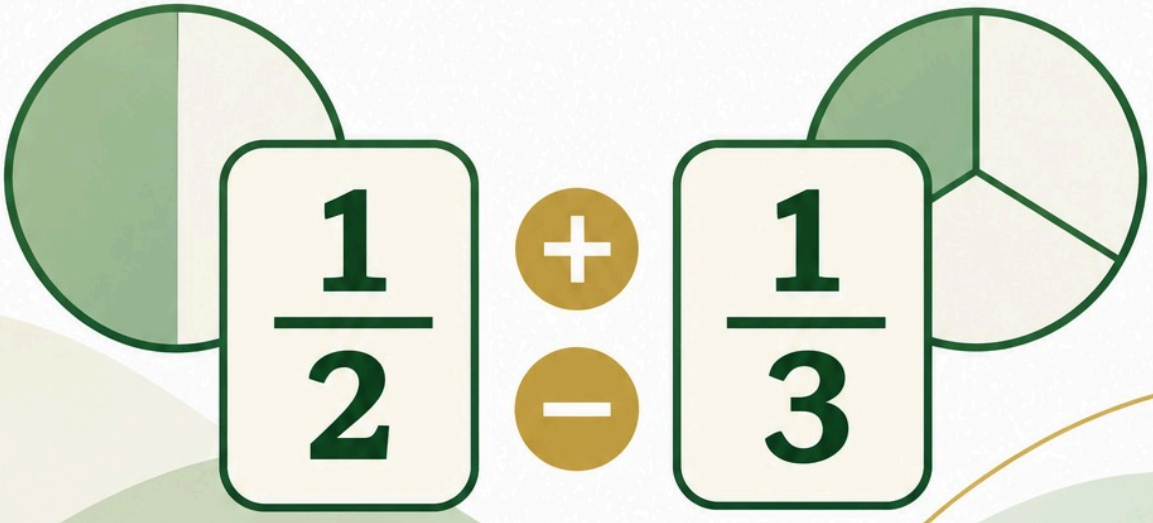
SCHOOL OF MATH

KID EINSTEINS

CLASS 19

Adding & Subtracting Fractions

Unlike Denominators



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Why Denominators Must Match

To add or subtract fractions, the parts must be the same size.

$$\frac{1}{2} + \frac{1}{3}$$



Halves and thirds are different-sized parts.

Rename using equal-sized parts.

$$\frac{1}{2} = \frac{3}{6}$$



$$\frac{1}{3} = \frac{2}{6}$$



$$\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$



Takeaway: The denominator names the size of the parts. Add the numerators; keep the common denominator.

Why denominators must match

Q1. Why is it wrong to add $\frac{1}{4} + \frac{1}{3}$ by adding the tops and the bottoms to get $\frac{2}{7}$? Find the correct sum.

Q2. Rename $\frac{1}{2}$ and $\frac{1}{5}$ as tenths. Then add them.

Q3. One bar is cut into fourths with 1 part shaded. Another bar is cut into eighths with 3 parts shaded. Rename so the parts are the same size, then find $\frac{1}{4} + \frac{3}{8}$.

Q4. Which pair can be added right away without renaming: $\frac{2}{5}$ and $\frac{3}{5}$, $\frac{1}{3}$ and $\frac{1}{6}$, or $\frac{3}{4}$ and $\frac{3}{8}$? Add that pair.

Q5. Lena says $\frac{1}{2} + \frac{1}{4} = \frac{2}{6}$. Explain her mistake and find the correct sum.

Finding a Common Denominator

Use multiples to find the least common denominator (LCD).

$$\frac{1}{4} + \frac{1}{6}$$

1. List multiples of each denominator

Multiples of 4: 4, 8, (12), 16, 20, 24

Multiples of 6: 6, (12), 18, 24, 30, 36

2. Choose the smallest shared multiple

LCD = 12

12 is the smallest number divisible by both 4 and 6.

3. Rename the fractions

Multiply top and bottom by 3.

$$\frac{1}{4} = \frac{3}{12}$$

Multiply top and bottom by 2.

$$\frac{1}{6} = \frac{2}{12}$$

$$\frac{3}{12} + \frac{2}{12} = \frac{5}{12}$$

A common denominator works. The least common denominator keeps the numbers smaller.

Try it: Find the LCD of 6 and 8.

Finding a common denominator

Q6. Find the least common denominator of 3 and 4.

Q7. Find the least common denominator of 4 and 10.

Q8. Find the least common denominator of 5 and 15. What shortcut can you use?

Q9. Find the least common denominator of 9 and 6.

Q10. Find the LCD for $\frac{1}{3} + \frac{1}{8}$. Then rename both fractions and add.

Building Equivalent Fractions

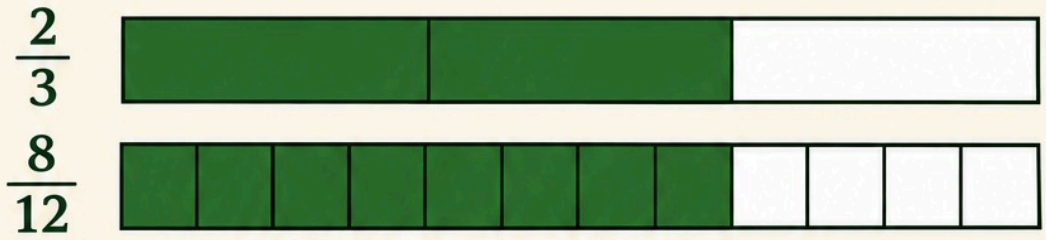
Different names. The same value.

Multiply the numerator AND denominator by the same number.

$$\frac{2}{3} = \frac{(2 \times 4)}{(3 \times 4)} = \frac{8}{12}$$

Multiplying by 4/4 is multiplying by 1, so the value stays the same.

See why it works



The shaded amount does not change.

Rename fractions with a common denominator

$\frac{2}{3}$ and $\frac{3}{4}$ → denominator 12

$$\frac{2}{3} = \frac{(2 \times 4)}{(3 \times 4)} = \frac{8}{12} \qquad \frac{3}{4} = \frac{(3 \times 3)}{(4 \times 3)} = \frac{9}{12}$$

Keep the value: multiply the top and bottom by the SAME nonzero number.

Try it: $\frac{3}{5} = \frac{?}{20}$ $\frac{5}{6} = \frac{?}{18}$

Building equivalent fractions

Q11. Fill in the missing numerator: $\frac{3}{4} = \frac{?}{16}$.

Q12. Fill in the missing numerator: $\frac{2}{7} = \frac{?}{21}$.

Q13. Fill in the missing denominator: $\frac{4}{9} = \frac{20}{?}$.

Q14. Rename $\frac{1}{6}$ and $\frac{3}{8}$ so that both have denominator 24.

Q15. Is $\frac{3}{5}$ equal to $\frac{9}{15}$? Is $\frac{2}{3}$ equal to $\frac{6}{8}$? Explain each.

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Adding Fractions with Unlike Denominators

$$\frac{3}{8} + \frac{1}{6}$$

Rename. Add. Simplify.

1. Find the least common denominator

Multiples of 8: 8, 16, **24**

Multiples of 6: 6, 12, 18, **24**

LCD = 24

2. Build equivalent fractions

$$\frac{3}{8} = \frac{3 \times 3}{8 \times 3} = \frac{9}{24}$$

$$\frac{1}{6} = \frac{1 \times 4}{6 \times 4} = \frac{4}{24}$$

3. Add the numerators

$$\frac{9}{24} + \frac{4}{24} = \frac{13}{24}$$

Keep the common denominator.

4. Check and simplify

13 and 24 have no common factor greater than 1.

Answer: $\frac{13}{24}$

Both fractions are positive, so the sum is greater than either fraction.

Add the numerators only after the denominators match.

Try it: $\frac{1}{4} + \frac{1}{6}$ $\frac{2}{5} + \frac{1}{3}$

Adding fractions with unlike denominators

Q16. Add: $\frac{1}{3} + \frac{1}{4}$.

Q17. Add: $\frac{2}{5} + \frac{1}{2}$.

Q18. Add: $\frac{1}{6} + \frac{3}{4}$.

Q19. Add and simplify: $\frac{3}{10} + \frac{1}{5}$.

Q20. Add. Write the answer as a mixed number: $\frac{5}{6} + \frac{3}{8}$.

Subtracting Fractions with Unlike Denominators

Rename. Subtract. Simplify.

$$\frac{5}{6} - \frac{1}{4}$$

1. Find the least common denominator.

Multiples of 6: 6, **12**

Multiples of 4: 4, 8, **12**

LCD = 12

2. Build equivalent fractions.

$$\frac{5}{6} = \frac{5 \times 2}{6 \times 2} = \frac{10}{12}$$

$$\frac{1}{4} = \frac{1 \times 3}{4 \times 3} = \frac{3}{12}$$

3. Subtract the numerators.

$$\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$$

Keep the common denominator.

4. Check and simplify.

7 and 12 have no common factor greater than 1.

Answer: $\frac{7}{12}$

Check: $\frac{7}{12} + \frac{3}{12} = \frac{10}{12} = \frac{5}{6}$

Subtract the numerators only after the denominators match.

Simplify when possible:

$$\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

Try it:

$$\frac{7}{8} - \frac{1}{3} \qquad \frac{5}{6} - \frac{1}{2}$$

Subtracting fractions with unlike denominators

Q21. Subtract: $\frac{3}{4} - \frac{1}{3}$.

Q22. Subtract: $\frac{4}{5} - \frac{1}{2}$.

Q23. Subtract and simplify: $\frac{7}{10} - \frac{1}{5}$.

Q24. Subtract: $\frac{5}{8} - \frac{1}{6}$.

Q25. Subtract and simplify: $\frac{11}{12} - \frac{3}{4}$.

Adding and Subtracting Mixed Numbers

Match the denominators. Regroup when needed.

ADDITION: Combine wholes and fractions

$$2\frac{1}{3} + 1\frac{3}{4}$$

Use denominator 12: $2\frac{4}{12} + 1\frac{9}{12}$

Add: $3\frac{13}{12}$

Regroup 13/12 as $1\frac{1}{12}$: $4\frac{1}{12}$

SUBTRACTION: Regroup one whole

$$4\frac{1}{4} - 2\frac{2}{3}$$

Use denominator 12: $4\frac{3}{12} - 2\frac{8}{12}$

Since 3/12 < 8/12, rename one whole: $4\frac{3}{12} = 3\frac{15}{12}$

Subtract: $3\frac{15}{12} - 2\frac{8}{12} = 1\frac{7}{12}$

One whole equals $\frac{12}{12}$.
Regrouping changes the form, not the value.

Try it:

$1\frac{2}{5} + 2\frac{1}{3}$ | $5\frac{1}{6} - 2\frac{3}{4}$

Adding and subtracting mixed numbers

Q26. Add: $1\frac{1}{2} + 2\frac{1}{3}$.

Q27. Add and regroup: $3\frac{3}{4} + 1\frac{1}{2}$.

Q28. Subtract: $5\frac{2}{3} - 2\frac{1}{4}$.

Q29. Subtract. Regroup if needed: $6\frac{1}{5} - 3\frac{1}{2}$.

Q30. Subtract from a whole number: $4 - 1\frac{3}{8}$.

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Solving Fraction Word Problems

Read. Choose an operation. Solve. Label.

ADD: Find the total

A recipe uses $\frac{2}{3}$ cup of milk and $\frac{3}{4}$ cup of water. How much liquid is used in all?



Combine the two amounts, so add.

$$\frac{2}{3} + \frac{3}{4} = \frac{8}{12} + \frac{9}{12} = \frac{17}{12} = 1\frac{5}{12}$$

$1\frac{5}{12}$ cups of liquid

SUBTRACT: Find what remains

A trail is $1\frac{1}{2}$ miles long. Maya has walked $\frac{5}{8}$ mile. How much farther must she walk?



Subtract the distance walked from the total.

$$1\frac{1}{2} - \frac{5}{8} = \frac{12}{8} - \frac{5}{8} = \frac{7}{8}$$

$\frac{7}{8}$ mile remaining

Before calculating, ask:

Am I combining amounts, finding what remains, or comparing amounts?

Use a common denominator and include the unit in your answer.

Try it

A ribbon is $\frac{5}{6}$ meter long. You cut off $\frac{1}{4}$ meter. How much ribbon remains?

Solving fraction word problems

Q31. Sam paints $\frac{1}{3}$ of a fence in the morning and $\frac{2}{5}$ of it in the afternoon. What fraction of the fence has he painted?

Q32. A jug holds $\frac{7}{8}$ gallon of juice. Mia pours out $\frac{1}{2}$ gallon. How much juice is left in the jug?

Q33. Ava runs $1\frac{2}{3}$ miles on Monday and $2\frac{1}{2}$ miles on Tuesday. How far does she run in all?

Q34. A board is $3\frac{1}{4}$ feet long. Dev cuts off $1\frac{5}{6}$ feet. How long is the piece that is left?

Q35. Leo reads for $\frac{3}{4}$ hour. Mia reads for $\frac{2}{5}$ hour. How much longer does Leo read than Mia?

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Mixed Practice

Estimate, Solve, and Check

$$\frac{7}{8} - \frac{1}{3}$$

1. Estimate

$\frac{7}{8}$ is close to 1, and $\frac{1}{3}$ is close to $\frac{1}{2}$.

$$1 - \frac{1}{2} = \frac{1}{2}$$

Expect an answer near $\frac{1}{2}$.

2. Solve

$$\frac{7}{8} - \frac{1}{3} = \frac{21}{24} - \frac{8}{24}$$

$$= \frac{13}{24}$$

3. Check

$\frac{13}{24}$ is close to $\frac{1}{2}$.

$$\frac{13}{24} + \frac{8}{24} = \frac{21}{24} = \frac{7}{8}$$

Your turn: Estimate → Solve → Check

1. $\frac{3}{4} + \frac{2}{5}$

2. $\frac{5}{6} - \frac{3}{8}$

3. $2\frac{1}{3} + 1\frac{5}{6}$

4. $4\frac{1}{4} - 2\frac{2}{3}$

✓ Common denominator | ✓ Simplest form | ✓ Reasonable answer

Mixed practice: estimate, solve, and check

Q36. Estimate, solve, and check: $\frac{5}{6} + \frac{1}{8}$.

Q37. Estimate, solve, and check: $\frac{9}{10} - \frac{2}{5}$.

Q38. Estimate, solve, and check: $\frac{2}{3} + \frac{4}{5}$.

Q39. Estimate, solve, and check: $3\frac{1}{6} - 1\frac{3}{4}$.

Q40. Noah says $\frac{7}{12} - \frac{1}{4} = \frac{6}{8}$. Use an estimate to show his answer is unreasonable, then find the correct difference.

Homework review

R1. Explain why $\frac{1}{3} + \frac{1}{6}$ is not $\frac{2}{9}$. Then find the correct sum.

R2. Find the least common denominator of 6 and 10.

R3. Fill in the missing numerators: $\frac{5}{6} = \frac{?}{24}$ and $\frac{3}{8} = \frac{?}{24}$.

R4. Add: $\frac{1}{4} + \frac{2}{3}$.

R5. Subtract: $\frac{5}{6} - \frac{2}{9}$.

Homework review (continued)

R6. Add and regroup: $2\frac{3}{5} + 1\frac{1}{2}$.

R7. Pia has $\frac{5}{6}$ cup of flour. She uses $\frac{2}{3}$ cup in a recipe. How much flour is left?

R8. Estimate, solve, and check: $\frac{4}{5} + \frac{3}{10}$.

Homework word problems

W1. In a garden, $\frac{1}{2}$ of the beds have tomatoes and $\frac{1}{6}$ of the beds have beans. What fraction of the beds have tomatoes or beans?

W2. A water bottle holds $\frac{9}{10}$ liter. Omar drinks $\frac{2}{5}$ liter. How much water is left?

W3. Rosa walks $\frac{3}{8}$ mile to the library and then $\frac{1}{4}$ mile to the park. How far does she walk in all?

W4. Ben mixes $\frac{2}{3}$ gallon of blue paint with $\frac{1}{5}$ gallon of white paint. How much paint does he have now?

W5. Kai practices piano for $1\frac{1}{4}$ hours on Saturday and $\frac{2}{3}$ hour on Sunday. How long does he practice over the weekend?

Homework word problems (continued)

W6. Zoe's plant is $5\frac{1}{2}$ inches tall. Eli's plant is $3\frac{3}{4}$ inches tall. How much taller is Zoe's plant?

W7. A bag holds 4 pounds of rice. The cook uses $1\frac{2}{3}$ pounds. How much rice is left in the bag?

W8. Maria has $2\frac{1}{2}$ cups of milk. She uses $\frac{3}{4}$ cup for pancakes and $1\frac{1}{3}$ cups for a smoothie. How much milk is left?

W9. On a track, Jada runs $\frac{3}{5}$ mile, walks $\frac{1}{4}$ mile, and then runs $\frac{1}{2}$ mile. How far does she travel in all?

W10. Jack eats $\frac{3}{8}$ of a pizza and Ana eats $\frac{1}{3}$ of the same pizza. What fraction of the pizza is left?

Homework word problems (continued)

W11. A recipe needs $2\frac{3}{4}$ cups of flour. Lucy has $1\frac{5}{6}$ cups. How much more flour does she need?

W12. A trail is 6 miles long. A family hikes $2\frac{3}{8}$ miles on the first day and $1\frac{3}{4}$ miles on the second day. How far do they still have to hike?